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Research Paper

Quantification of litter in cities using a smartphone application and citizen science in conjunction with deep learning-based image processing

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ABSTRACT

Cities are a major source of litter pollution. Determination of the abundance and composition of plastic litter in cities is imperative for effective pollution management, environmental protection, and sustainable urban development. Therefore, here, a multidisciplinary approach to quantify and classify the abundance of litter in urban environments is proposed. In the present study, litter data collection was integrated via the Pirika smartphone application and conducted image analysis based on deep learning. Pirika was launched in May 2018 and, to date, has collected approximately one million images. Visual classification revealed that the most common types of litter were cans, plastic bags, plastic bottles, cigarette butts, cigarette boxes, and sanitary masks, in that order. The top six categories accounted for approximately 80 % of the total, whereas the top three categories accounted for more than 60 % of the total imaged litter. A deep-learning image processing algorithm was developed to automatically identify the top six litter categories. Both precision and recall derived from the model were higher than 75 %, enabling proper litter categorization. The quantity of litter derived from automated image processing was also plotted on a map using location data acquired concurrently with the images by the smartphone application. Conclusively, this study demonstrates that citizen science supported by smartphone applications and deep learning-based image processing can enable the visualization, quantification, and characterization of street litter in cities.

1. Introduction

Marine pollution caused by plastic waste is a serious global environmental problem. Plastics entering the ocean are not directly dumped into the ocean but are primarily domestic waste discharged from cities into the ocean via rivers (Lebreton et al., 2017; Morales-Caselles et al., 2021). According to Jambeck et al. (2015), of the 275 million tons of plastic waste produced in 2010, 31.9 million tons were released into the environment owing to mismanagement, and 4.8–12 million tons of this quantity were discharged into the ocean. To date, approximately 25 million tons of plastic waste, which is only 5 % of the total amount of plastic waste, has been discharged into the ocean (Isobe and Iwasaki, 2022). In other words, the remaining 95 % is lost on land and may partially end up as marine plastic waste.

At the 2019 Osaka G20 Summit, the Osaka Blue Ocean Vision set forth the goal of zero additional marine plastic pollution by 2050 (https://g20mpl.org). To achieve this, it is important to determine the abundance and composition of plastic litter in the city, which is a major source of marine plastic pollution. In addition, the location of the discarded plastic that may remain on land must be clarified. Otherwise, it is difficult to determine to what extent the reduction of land plastic content will reduce additional ocean pollution and accurately assess its current and future environmental effects.

Many researchers have conducted beach observations to investigate litter abundance and composition (Ribic, 1998; Derraik, 2002; Kako et al., 2010). For example, Kako et al. (2010) and Ambrose et al. (2019)

Abbreviations: YOLOv5, You Only Look Once; UAV, Uncrewed aerial vehicles.

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quantified litter washed ashore on Japanese and Bahamian beaches, respectively, based on data collected using citizen science. Most of these studies have been conducted manually. However, recently, methods combining aerial beach observations with uncrewed aerial vehicles (UAVs) and machine learning-based image analysis have been developed to accurately quantify plastic litter (e.g., Martin et al., 2018, 2021; Fallati et al., 2019; Kako et al., 2020; Gonçalves et al., 2022; Hidaka et al., 2022; Jia et al., 2023). These studies have demonstrated the effectiveness of plastic litter quantification based on image analysis and UAV surveys to accurately monitor large areas. However, it is difficult to apply bulk UAV observations in cities with high human traffic and buildings owing to the risk of crashes and collisions. Moreover, many countries do not allow drone flights in urban areas because of legislation rules.

Quantitative manual surveys of street litter have also been carried out in various parts of the world, including New York, (Van Ryzin et al., 2008) and Los Angeles in the USA (Ladin-Sienne, 2017), and Granada in Spain (Silva et al., 2018). These studies were conducted to investigate street cleanliness, with the assumption that this may prevent the illegal dumping of litter. Polluted streets create a lack of hesitation on illegal dumping (the Broken windows theory, e.g., Keizer et al., 2008). Liedu et al. (2022) and Tramoy et al. (2022) investigated the abundance of macro-litter on an 800-m roadway in Nantes, France, to clarify the source of the macro-litter. Tesfaldet et al. (2022) conducted a 42-day observational study in three streets in Bangkok, Thailand, to assess the illegal dumping of sanitary masks during the coronavirus disease 2019 pandemic on urban pollution.

Previous studies have also attempted to quantify plastic waste through citizen science. Quantification of litter in the UK and German cities carried out by Winton et al. (2023) and Kiessling et al. (2023), respectively, was based on citizen science campaigns conducted several times a year. Scardino et al. (2022) conducted a study using citizen science based on a smartphone application and quantified plastic litter on Italian beaches and coastal areas, although their survey involved the collection of only approximately 100 images. Their study introduced novel ideas but relied on manual collection and counting to quantify litter information, which makes continuous observation and quantification difficult owing to high cost and time consumption.

Recent advancements in litter survey methods have yielded alternatives to manual observation for assessing illegal dumping and cleanliness in cities using cameras mounted on mobile stations (Burger et al., 2018; Li et al., 2018), garbage trucks (Ping et al., 2020), and bicycles (Hagura et al., 2023). These studies have demonstrated the effectiveness of using machine learning and deep learning techniques based on Convolutional Neural Network (CNN), which have made remarkable progress in quantifying street litter in recent years. However, the observation range of these studies is limited to a few city blocks or few case studies owing to cost-prohibitive and other issues, and challenges persist in collecting a wide range of data sequentially and continuously.

A smartphone application for citizen science is useful by contributing to permanent and extensive data collection (from the city to the country and the world). The importance of citizen science in the implementation of long-term monitoring of plastic litter has been suggested by Shruti et al. (2022). An observation system using a smartphone application can be used by anyone without special skills for accuracy, for citizens to access free of charge, and for operationally collecting a large amount of data. In addition, modern smartphone cameras have a resolution of approximately 12–48 megapixels, which is sufficient for capturing quality images that can be analyzed to classify and quantify litter using image processing. Moreover, based on the location data recorded in smartphones, it is possible to easily identify where the street litter was photographed. This approach is suitable for long-term data collection, as litter pictures can be captured whenever citizens encounter litter or wish to take a photograph. In line with this approach, there have been attempts to visualize litter in cities, riverside, and coastal regions using smartphone applications in some projects such as OpenLitterMap (https://openlittermap.com/), PLANET PATROL (https://planetpatrol.co), and Pirika (https://corp.pirika.org/en/sns-pirika/). However, the automatic processing of images has not yet led to the classification and quantification of these systems. In this study, images derived from the smartphone application Pirika downloaded by citizens with no strict protocol were used to quantify litter in cities. Data collection through citizen science with Pirika is employed worldwide (see section 2). Therefore, the quantification of large amounts of data is possible, followed by subsequent processing of these images automatically.

The aim of the present study was to investigate whether it is possible to visualize the location and abundance of street litter (the number of litter pieces) by integrating litter data collection using Pirika (citizen science) with image analysis based on deep learning. Although some previous studies have estimated the abundance of litter through image analysis based on deep learning (Burger et al., 2018; Li et al., 2018; Ping et al., 2020; Hagura et al., 2023), to the best of our knowledge, the present study is the first to quantify mismanaged plastic waste littered in cities in a framework of citizen science supported by deep learning-based image analysis. To encourage non-expert citizen participation in collecting a large amount of image data, this study considerably omitted strict rules. Unexpected human errors are unavoidable when collecting large amounts of image data. Therefore, data quality was ensured by deep learning. Confirming the suitability of this approach was also the purpose of this study. Furthermore, this study serves as an initial step toward quantifying and categorizing, and in turn, clarifying the plastic waste flow from cities to oceans via rivers.

2. Material and methods

2.1. Observation and quantification concepts

A flowchart illustrating the process of data collection, quantification, classification, and depiction of litter abundance in cities is presented in Fig. 1. This study focused on visualizing the abundance of litter in cities, as indicated by the blue route outlined in Fig. 1. Pirika (i.e., citizen science with a smartphone app) enables the long-term collection of a large amount of litter data with location information in a wide range, which was not achieved in previous studies. It is almost impossible to manually classify such global data and extract information on urban litter from these images continuously. To address this challenge, it is imperative to develop a system capable of automatically extracting object information from images and tallying the quantity of each object. This automated approach is essential for mapping the distribution and abundance of litter. In this section, the smartphone app and deep learning model used in this study are described.

2.2. Smartphone application "Pirika"

The smartphone application "Pirika" used in this study runs on both iOS and Android. Both operating system versions can be downloaded for free from the App Store and Google Play. Pirika is available as a Social Networking Service (SNS) application available in more than 110 countries worldwide (the country with the most downloads is Japan). The application enables users to upload images captured in cities with location information to the server to archive them. An example of an image submitted via Pirika is shown in Fig. 1. When uploading images, users can add their comments on them, as well as interact and share information with other users in the SNS framework. Data archiving has been ongoing since May 2018. The sole guideline for gathering litter data in urban areas is to photograph any litter encountered using Pirika and upload it. If possible, the litter should be disposed-off. Our projects refrained from imposing any requirements on citizens (Pirika users). This approach was adopted for collecting a range of data over a long period.

Numerous images have been submitted from all over Japan through Pirika, as depicted in the map illustrating the average of submitted

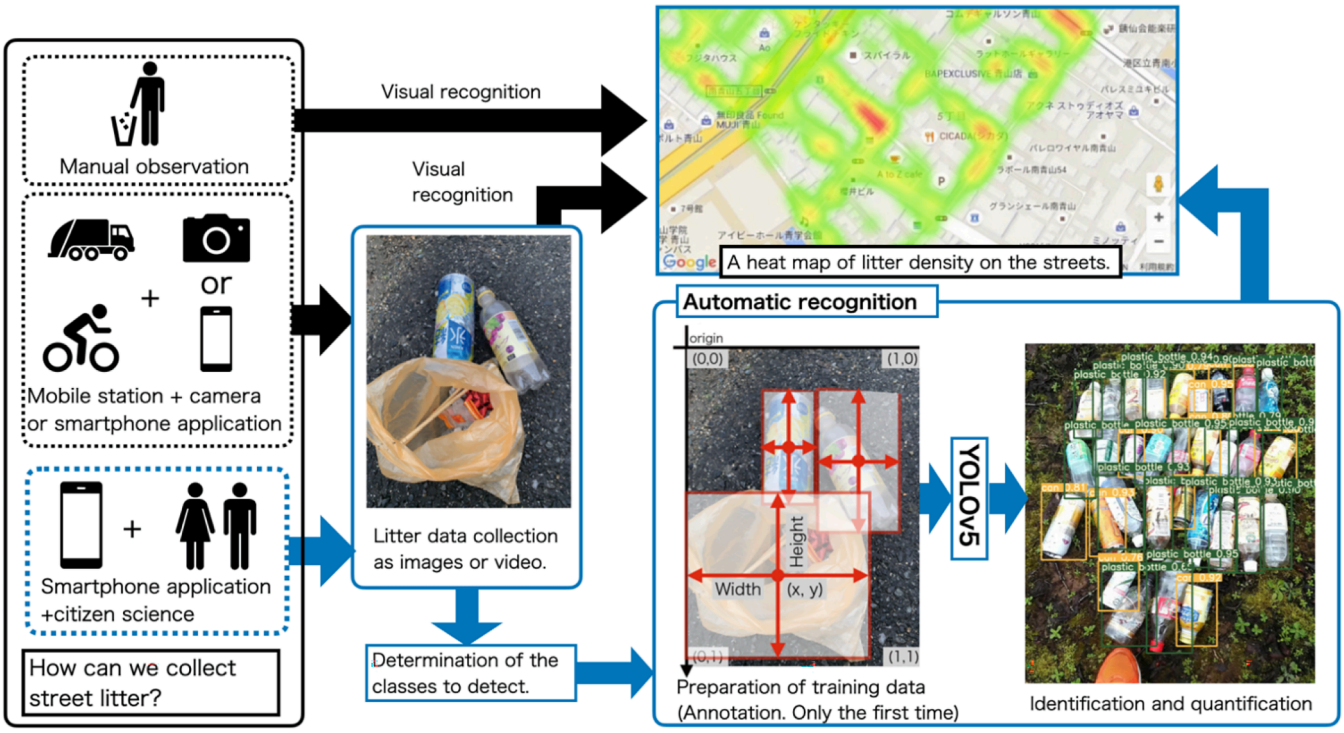


Fig. 1. Flowchart depicting the process of generating a heatmap indicating litter density in cities to investigate the cleanliness, predominant types of litter, litter reservoirs, and related factors. The methods involved are indicated in blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

images aggregated over every 20-km grid during May 2022 (Fig. 2a). The map indicates that more submissions come from urban areas such as Tokyo, Nagoya, and Osaka than from the countryside and, thus, the litter data collected by Pirika are representative of urban litter. This characteristic was independent of both year and season (not shown). The number of submissions increased over time (Fig. 2b), averaging approximately 37,000 submissions per month since 2022. To date, approximately one million images have been submitted, and sufficient images have been accumulated for training image processing methods based on deep learning.

2.3. Object detection and quantification with deep learning

An existing and well-validated method was used to make the system accessible to anyone to implement anywhere provided that suitable training data were available for their location, enabling the estimation of litter in cities. The deep learning model used in this study for object detection in the images was You Only Look Once version 5 (YOLOv5). YOLOv5 based on CNN was released in June 2020 and can be downloaded for free by anyone from GitHub (Jocher, 2020; https://github.com/ultralytics/yolov5). For example, Song et al. (2021), Teng et al. (2022), and Chen and Zhu (2024) have already used YOLO for the quantification of beach and water surface litter and demonstrated its

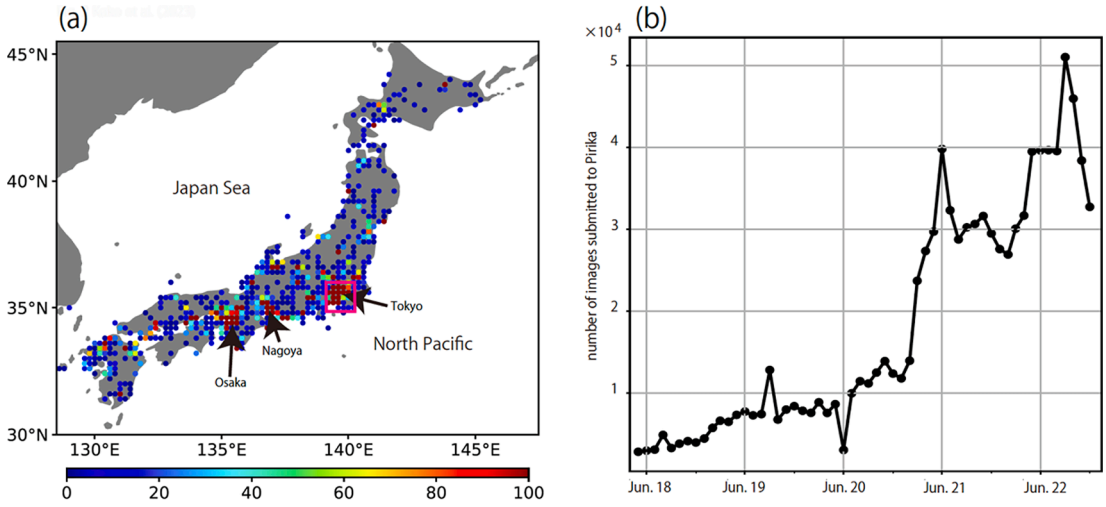


Fig. 2. (a) Average number of images submitted on September 2022 aggregated per 20-km grid. The pink square indicates the Kanto region shown in Fig. 5. Time series of the number of images submitted to Pirika between May 2018 and December 2022. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

effectiveness in detecting litter in images.

Although YOLOv5, similar to its earlier versions (ver.1–4), detects and classifies objects in an image using bounding boxes, it implements PyThorch and not Darknet, which was used in the earlier versions. In addition, YOLOv5 has made improvements to data augmentation methods introduced in YOLOv4, namely, Mosaic (a method of combining several images) and MixUP (a method of averaging two images). For more detailed specifications and algorithms of YOLOv5, refer to the Ultralytics LLC website (https://ultralytics.com).

Before training the model, it was necessary to determine the number of types (original products such as plastic bottles and bags) of litter in cities to be classified. Therefore, 637 images submitted to Pirika were randomly selected and different types of litter were manually counted. Using the results of these visual observations, the target litter types that were frequently detected in cities (see Section 3) were determined. Annotation to create the training data was performed using LabelImg, which can be downloaded free of charge from GitHub (https://github.com/tzutalin/labelImg). For annotation, YOLOv5 requires the data shown in the second image from left in Fig. 1. Thus, the coordinates of the center of the object and the width and height from it are required. However, using LabelImg, such data can be easily obtained by outlining the object using a graphical user interface. The dataset was further expanded using Mosaic and MixUP, as described above. In addition, fine-tuning with Microsoft Common Object in Context (MSCOCO) data was used to improve the training efficiency. COCO is available free of charge as a dataset of over 200,000 images and annotation combinations; it can be used for object detection and segmentation, among others. It is useful for improving the training efficiency of YOLOv5. Additionally, in this study, a system to compute the amount of litter present in each image was constructed. To this end, YOLOv5 was used to detect objects and identify litter type with location information for each image derived from Pirika; the information was stored in a separate file for quantification (number of litters) for each location.

To validate the accuracy of YOLOv5, precision, recall, and F-score were calculated for each type of litter using the following equations:

$$Precision = \frac{Number\ of\ estimated\ litter\ that\ were\ correct}{Number\ of\ estimated\ litter} \quad (1)$$

$$Recall = \frac{Number\ of\ estimated\ litter}{Number\ of\ actuall\ litter}, \quad (2)$$

and

$$F-score = 2\cdot\frac{precision\cdot recall}{precision + recall}. \quad (3)$$

Precision is an indicator of how much of what YOLOv5 determined to be litter was actually litter. Recall is an indicator of how much of the actual litter in the image was detected by YOLOv5. F-score is the harmonic mean of precision and recall. A value closer to 1 signifies higher precision and recall, indicating the most efficient and balanced machine/deep learning model. This is a commonly used metric because of the trade-off relationship between precision and recall (e.g., when attempting to increase the recall to ensure that there are no missed objects, the number of misdetections tends to increase and the precision tends to decrease).

3. Results

3.1. Observation results of citizen science using Pirika

Visual observations were conducted to investigate the types of images submitted. From all images, 637 images were randomly selected and categorized into images with and without litter. Approximately 40 % of the images were not related to litter in the city and included images of meals, landscapes, and dogs and other pets. Of the remaining images, 195 (approximately 30 % of the total) were images of litter in the city but were not suitable for training the deep learning model and quantifying the litter, as the objects were out of focus or the objects could not be visually classified because of inappropriate brightness or angle of view. Litter in the city was clearly captured in 189 of the 637 images. Thus, even without specific skills or guidance, 30 % of the data were useful for visualizing and quantifying litter in the city. Currently, an estimated 30,000 images are submitted every month. To date, approximately one million images have been submitted, of which approximately 300,000 images may specifically capture urban litter, which can be used for the quantification and classification of litter in cities. Therefore, continuous data collection across a large area has been achieved through Pirika.

To quantify the litter in cities, YOLOv5 was trained using images captured by Pirika. This needed an understanding of the characteristics and composition of the litter derived from the submitted images. Therefore, the type of litter in the remaining 189 images was investigated via visual observation. The types of litter in the descending order of the number of data were cans, plastic bags, plastic bottles, cigarette butts, cigarette boxes, and sanitary masks, with the top six types accounting for 78 % and the top three types for 64 % of the total 189 litter images (Fig. 3). The food packaging category was more prevalent than sanitary masks (Fig. 3). However, there was no further subdivision into coffee cups, straws, caps of plastic bottles, and snack bags, each of which was less common than sanitary masks. When grouped by similar shapes, masks were the sixth most common litter type in cities.

3.2. Results of YOLOv5 training

Considering the results, two deep learning models were constructed. The first was capable of detecting the top three litter types and the second was capable of classifying the top six litter types. These two models enabled the assessment of the change in the accuracy of the deep learning model with an increasing number of detectable litter types. To construct these deep learning models that could detect the top three and six types of litter, training data were created using 1219 and 2054 images, respectively, using LabelImg (see Section 2). The total number of annotations was 1532 and 2757, respectively. Of these, 70 % were used for training and 30 % were used for validation without cross-validation. The models were trained for over 300 epochs under these conditions by a computer equipped with a graphical processing unit (GPU), NVIDIA GeForce RTX 3090. As a result, the training time for YOLOv5 was approximately 5 h. The time required for object identification and classification using the trained model is approximately 0.02 s per image.

3.3. Results of image detection and quantification using YOLOv5

Fig. 4 shows an example showcasing object detection and classification of images using YOLOv5. YOLOv5 allows simultaneous object detection and classification (Song et al., 2021; Teng et al., 2022; and Chen and Zhu, 2024). The object is surrounded by a bounding box, and the type of litter is displayed in the top left-hand corner. As shown in Fig. 4, although some plastic bags often contained litter, they were classified as plastic bags without considering their contents.

To verify the accuracy of YOLOv5, 756 images that were not used to train YOLOv5 were extracted. The number of the six types of litter was visually counted, and the results were then compared with those obtained using YOLOv5 by computing precision, recall, and F-score described in section 2. These values of our model were evaluated using images with clearly observed litter features. Thus, images that were out of focus, of low brightness, or of low resolution were excluded. Table 1 shows that the three-type classification model achieved values, precision, and recall rates ranging from 80 % to 90 %, whereas the six-type classification model achieved values over 75 %. The F-score values of both models were more than 80 %. These values are comparable to or slightly higher than those reported in previous studies (see peer-

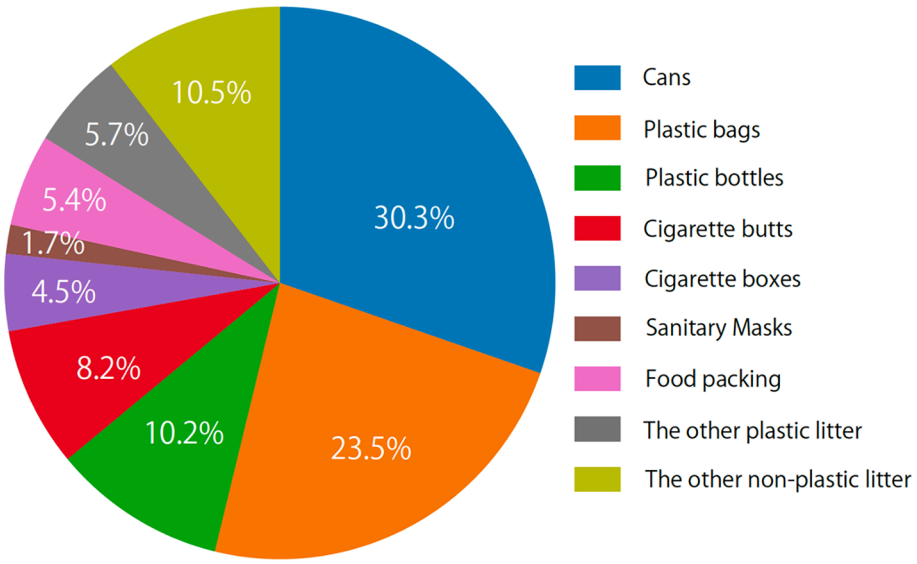


Fig. 3. Composition of street litter based on the visual classification of images submitted to Pirika. This pie chart is the result of visually classifying 189 of the 637 randomly extracted images in which street litter was clearly captured.



Fig. 4. Examples of image classification performed by YOLOv5 trained using images submitted through Pirika. Objects detected by YOLOv5 are surrounded by bounding boxes, the type of which is indicated in the top left-hand corner of the bounding box. If the number displayed in the bounding boxes is close to 1 (0), it indicates a high probability that YOLOv5 recognizes it as an object (background). In this study, only those with a value above 0.25 were used.

reviewed article by Jia et al., 2023). There were no significant differences in the values for cans, plastic bottles, and plastic bags (Table 1), between the models. However, the precision and recall of cigarette boxes in the six-type classification model were less than those of the other litter types. The possible causes of misidentification in our model are explored in the next section. In summary, these results suggest that with sufficient training data, more classifications can be made without compromising accuracy. To achieve this, it is necessary to continue collecting litter data that can effectively be used to train image processing methods through citizen science.

Based on the results of YOLOv5, scatter plots indicating the abundance of each litter type were established (Fig. 5). Specifically, the abundance of litter in the Kanto region around Tokyo, Japan, with a high number of submissions compared with other regions, was plotted on the map (Fig. 2a). Even within the heavily populated Kanto region (housing approximately 43 million people), most of the submissions originated from the suburbs of Tokyo, with a large population, whereas only a few submissions were obtained from rural areas such as Chiba Prefecture. There were also areas where image contributions were crowded and formed a distinct linear pattern, as indicated in the area enclosed in the red box. This region corresponds to the area around the Arakawa River, and the results suggest that there is a lot of litter around the river, indicating the accumulation of litter discharged from cities along the riverside. Implementing pollution control measures could potentially mitigate litter discharge into the ocean via rivers.

4. Discussion

4.1. Importance and potential of citizen science

Citizen science, facilitated through smartphone applications, is important for collecting large amounts of data across wide areas without any campaign and enhancing the quality of collected data using machine learning and deep learning algorithms (Van Ryzin et al., 2008; Burger

Table 1
Comparison of the results of the three- and six-type classification models with those of visual observation. Values in parentheses are the results of the three-type classification model.

	True value	Estimation	True positive	Precision	Recall	F-Sore
Cans	265 (236)	307 (256)	252 (220)	0.82 (0.86)	0.95 (0.93)	0.88 (0.90)
Plastic bags	272 (242)	273 (247)	252 (232)	0.92 (0.94)	0.93 (0.96)	0.93 (0.93)
Plastic bottles	192 (171)	190 (170)	175 (154)	0.92 (0.91)	0.91 (0.90)	0.92 (0.91)
Cigarette butts	288	240	216	0.90	0.75	0.82
Cigarette boxes	54	62	47	0.76	0.87	0.81
Sanitary masks	101	88	80	0.91		0.79
Standard error	–	–	–	0.02 (0.02)	0.03 (0.01)	0.02 (0.01)

et al., 2018; Ladin-Sienne, 2017; Li et al., 2018; Silva et al., 2018; Ping et al., 2020; Hagura et al., 2023; Tesfaldet et al., 2022). This study adopted a multidisciplinary approach, utilizing the Pirika smartphone application for litter data collection in urban environments, coupled with image analysis through a deep learning algorithm, to quantify and classify litter abundance. Our findings support the possibility of creating a global database of urban litter through citizen science, requiring only smartphones and willing participants. Notably, data were successfully collected from various regions in Japan for over 4 years without a campaign, surpassing previous efforts (Kako et al., 2010; Ambrose et al., 2019; Winton et al., 2023; Kiessling et al., 2023; Scardino et al., 2022; Tesfaldet et al., 2022). It demonstrates that by engaging the public, awareness of environmental issues such as pollution caused by litter can be improved. It also demonstrates the possibility to visualize the localization and abundance of litter in cities based on data collected by citizen science, as well as, objectively and accurately plotting the abundance of targeted litter on a map.

In terms of statistical value (Table 1), the image analysis method based on YOLOv5 constructed in this study is not inferior to the methods used in previous studies (see peer-reviewed article by Jia et al., 2023). Nevertheless, challenges remain in terms of the global deployment of the system developed in this study. For example, the number of litter types identified in this study was determined based on visual surveys, with the top six types accounting for more than 80 % of the total. However, these results are limited to Japan and may not be directly applicable to other countries owing to variations in litter composition. Given the diverse nature of litter in cities worldwide, the construction of a citywide litter classification system that can be applied globally within a short period is challenging. For instance, the litter composition in Medellín, Colombia, reported by Ulloa-Torrealba et al. (2023), is different from that in Japan captured by Pirika in this study. Therefore, to visualize street litter around the world automatically, it is important to build a system to collect images that can be used for training an image-processing model. In response, as this study has demonstrated, citizen science is always useful. Understanding the approximate composition of litter in a specific city is required to develop classification and quantification models using image processing methods based on machine learning. Thus, when constructing models, it is necessary to decide in advance what is to be detected. The deployment of the observation network with citizen science would be one of the major steps in clarifying the relationship between cities and rivers, and rivers and oceans, in the context of marine plastic pollution.

4.2. *Limitations, improvements and future directions*

There are several limitations to our current approach. The first limitation is the occurrence of false positives in the classification model, particularly because of similarities in the shape of different litter items. To address the factors causing false positives (i.e., misdetection) in our six-type classification model, a confusion matrix was computed (Fig. 6).

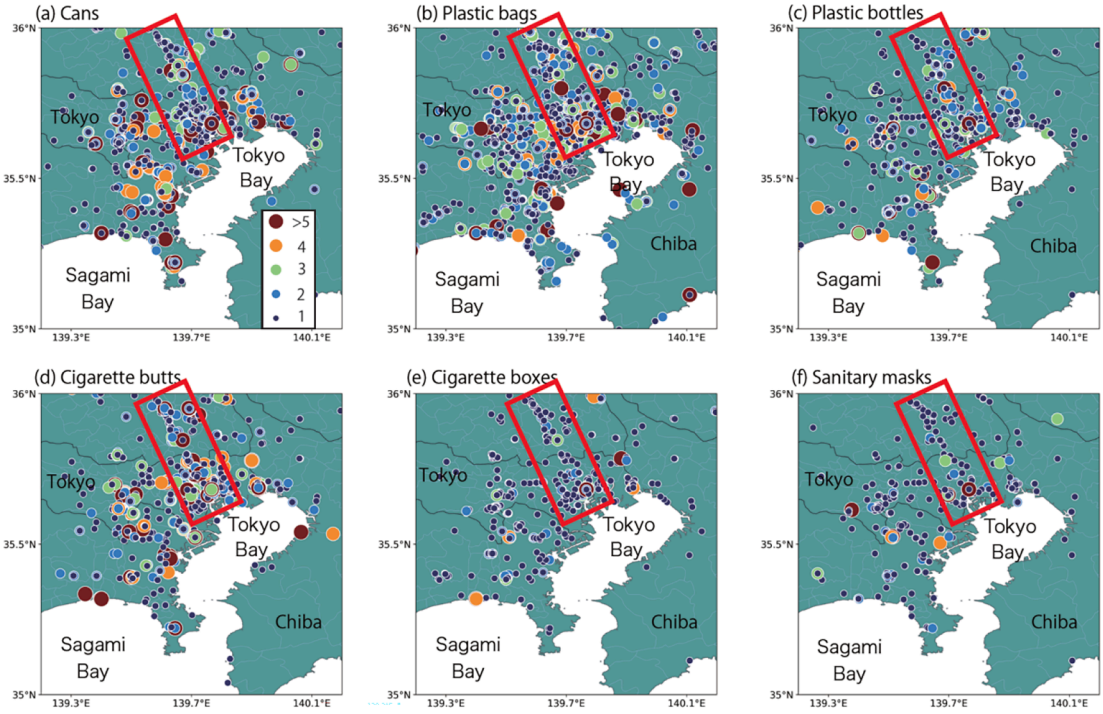


Fig. 5. Horizontal distribution of the abundance of litter estimated from YOLOv5 in May 2022. (a) Cans, (b) plastic bottles, (c) plastic bags, (d) cigarette butts, (e) cigarette boxes, and (f) sanitary masks. The area around the Arakawa River is indicated with by red boxes. The numbers in the legend in the upper left panel indicate the number of each object. All datapoints greater than five are represented by circles of uniform size. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The classification results from YOLOv5 and visual observation were then compared. The analysis revealed instances of misclassification, particularly where items shared similar shapes, such as plastic bottles being erroneously identified as cans and vice versa, representing a common challenge in deep learning models using bounding boxes. For example, while the model correctly identified 82 % of cans, aligning with visual observation, it misidentified 7 % of cans as plastic bottles. Notably, the total percentage may not add up to 100 % owing to instances where the background is detected as litter. Therefore, there is no significant bias towards any single false positive category. However, a relatively high proportion of plastic bottles was misidentified as cans and cans as cigarette boxes. Song et al. (2021) and Hagura et al. (2023) also noted similar limitations of deep learning models using bounding boxes. Thus, in cases where many objects of similar shape are collected, errors in composition may persist despite accurate counting of total waste items. There is a need for improved training data preparation for more accurate classification of such similarly shaped litter to address the challenges.

In addition, YOLOv5 has difficulty extracting fine objects, as also highlighted by Song et al. (2021), who demonstrated that YOLOv5 cannot accurately distinguish between small shells and plastic litter on beaches. This can be overcome by object detection using HRNet (Sun et al., 2019; Wang et al., 2021), recognized by Hidaka et al. (2022) for its ability to extract fine features. Notably, while HRNet offers enhanced detection capabilities, its implementation comes with a trade-off, sacrificing computation speed for improved precision.

The performance of the models was also affected by low-quality images. As Pirika is a free application, some images submitted were expected to be not suitable for quantifying urban waste. Low-quality images affect automated interpretation. Such images shown in Fig. 7 are of low brightness, have small object sizes compared to the background ground, or are out of focus, challenging object detection. The low-quality images submitted are probably because of the lack of guidance when capturing images using Pirika. Therefore, it is imperative to introduce a feature in Pirika that educates users on the kind of images required and how they can be captured. Future versions of the application could incorporate guidelines into the smartphone app's camera interphase to encourage users to capture photos suitable for surveying litter in cities to accumulate clear street litter images. Empirical studies to examine whether this is effective would also be necessary in the future.

The number of participants in citizen science increases gradually, which may potentially affect the spatiotemporal variations in the amount of urban litter collected. The present study aims to address this phenomenon in a future study and investigate ways to mitigate potential adverse effects. Additionally, the absence of submissions in certain locations via Pirika does not necessarily indicate zero waste. It rather signifies unknown litter abundance in those areas. Therefore, citizen science alone may not be sufficient for obtaining data for specific locations. This problem can be solved by organizing targeted campaigns to gather data for desired locations using a smartphone application with image analysis. Moreover, in recent years, attempts have been made to accurately estimate the amount of litter on beaches from insufficient data using a hierarchical Bayesian model (Matsuba et al., 2023). The application of such a statistical model to our data may enable the estimation of the spatiotemporal variability of street litter without any bias associated with participant numbers.

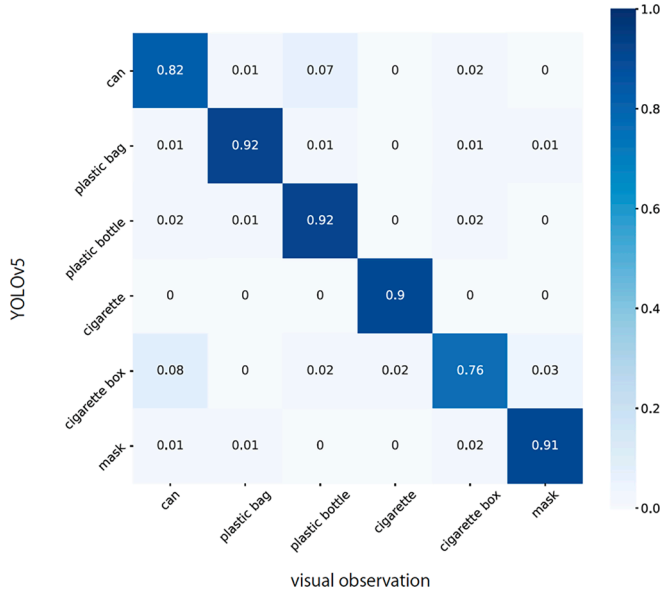


Fig. 6. Confusion matrix derived from a six-type classification model. All values are divided by the true value to obtain dimensionless units.

4.3. Managerial, political, and theoretical implications

There are various managerial, political, and theoretical implications of our study findings. First, in terms of managerial implications, this study shows that it is possible to collect large amounts of data using citizen science and to ensure its quality using deep leaning-based image processing. If the smartphone application as an observation tool can be made widely available, in the future, it will be possible to visualize the level of pollution in cities objectively and in real-time. Long-term and continuous observation, which has been difficult until now, will also become possible. This allows visualization of where and what kind of waste is driving pollution in cities. Second, in terms of political implications, if continuous monitoring in the cities described above is achieved, it will be possible to formulate policies on where and how waste can be managed to improve the level of pollution in cities. Although the use of disposable products such as plastic bags and straws is currently restricted and biodegradable products have been introduced, it is difficult to objectively assess their effect on improvements in the level of pollution in cities. However, the system presented in this study allows for continuous monitoring, which makes it possible to objectively know the effects of policies in reducing single-use plastic and introducing new plastic-alternative products. This will enable policymakers to demonstrate to the public the effectiveness of their policies based on scientific evidence. Once the effectiveness of scientifically formulated policies is demonstrated, citizens will better understand their continued enforcement leading to more active participation. Finally, concerning theoretical implications, cities are a source of marine plastic pollution. The global environmental flow of plastic waste from the vast terrestrial sources to the ocean via rivers, where the waste is crushed into microplastics and then deposited in beaches, seabeds, and the biosphere can be understood only when plastic litter can be quantified. Plastic litter is currently observed in rivers and beaches at various units and resolutions using various remote sensing methods such as UAV, webcam, and aircraft (Kako et al., 2018, 2020; Martin et al., 2018, 2021; Moy et al., 2018; Andriolo et al., 2020a, 2020b, 2024; Kataoka and Nihei, 2020; Gonçalves et al., 2022). In the future, it will be necessary to discuss standardization, harmonization, and multiscale coupling methods to combine them, which is necessary to elucidate the flow of plastic waste.

5. Conclusion

By collecting data over 4 years throughout Japan, the present study demonstrated and visualized the pervasive leakage of various disposable products, including beverages, cigarettes, and sanitary masks, into cities. This was made possible by citizen science and the accurate estimation of the abundance and composition of litter in the cities using deep-learning-based image analysis. Such detailed litter composition studies may have implications in reducing the costs of the surveys and improving citizens' environmental literacy. Our findings may enable the formulation of effective environmental policies, as well as, the



Fig. 7. Example of low-quality images unsuitable for object detection and quantification submitted to our server through Pirika.

assessment of the effectiveness of policy implementation through continuous monitoring by citizens. For instance, identifying primary sources of urban pollution through our approach could lead to global efforts to reduce the amount of plastic waste flowing from the cities into the ocean via the rivers. Our method would also be ideal for campaigns targeting specific litter in designated urban areas. Nevertheless, continuous development of new and improved observation methods and their integration with the latest image analysis techniques in future studies are necessary to enhance the efficient and effective control of plastic waste flow.

CRediT authorship contribution statement

Shin'ichiro Kako: Writing – original draft, Visualization, Software, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization. **Ryunosuke Muroya:** Writing – review & editing, Visualization, Validation, Investigation, Formal analysis. **Daisuke Matsuoka:** Writing – review & editing, Validation, Investigation. **Atsuhiko Isobe:** Writing – review & editing, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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